

MAGeCK Count Report

```
# define the comparison_name here; for example,  
# comparison_name='demo'  
comparison_name='R1formattedH1'
```

Note: if the sample labels are too long, they may cause issues in some figures. If this is the case, set the following variable to “TRUE”, and a numbered label will be shown instead of the original label.

```
REPLACE_LABEL=FALSE
```

Preprocessing

Reading input files. If any of these files are problematic, an error message will be shown below.

```
cstable=read.table(count_summary_file,header = T,as.is = T)  
nc_table=read.table(normalized_cnt_file,header = T,as.is = T)
```

Summary

```
## [1] "/Users/carolinestokes/Desktop/H1_subpooledlib_sequencing"
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
## Warning: package 'tidyr' was built under R version 4.2.3
```

```
## Warning: package 'readr' was built under R version 4.2.3
```

```
## Warning: package 'dplyr' was built under R version 4.2.3
```

```
## Warning: package 'stringr' was built under R version 4.2.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr   1.5.1
```

```
## v ggplot2    3.5.1      v tibble    3.2.1
```

```
## v lubridate  1.9.4      v tidyr     1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

The summary of the count command is as follows.

Table 1: Count command summary

File	Label	Reads	Mapped	Percentage	TotalsgRNAs	Zerocounts	GiniIndex
21062D-05-01_S1_L001_R1_001.fastq	L1	47454700	40510927	0.8537	13008	38	0.04939

The meanings of the columns are as follows.

- File: The filename of fastq file;
- Label: Assigned label;
- Reads: The total read count in the fastq file;
- Mapped: Reads that can be mapped to gRNA library;
- Percentage: The percentage of mapped reads;
- TotalsgRNAs: The number of sgRNAs in the library;
- ZeroCounts: The number of sgRNA with 0 read counts;
- GiniIndex: The Gini Index of the read count distribution. Gini index can be used to measure the evenness of the read counts, and a smaller value means a more even distribution of the read counts.

If `-day0label` and `-gmt-file` options are provided, the following metrics will display the degree of negative selections of essential genes (provided by `-gmt-file`).

Table 2: Count command summary

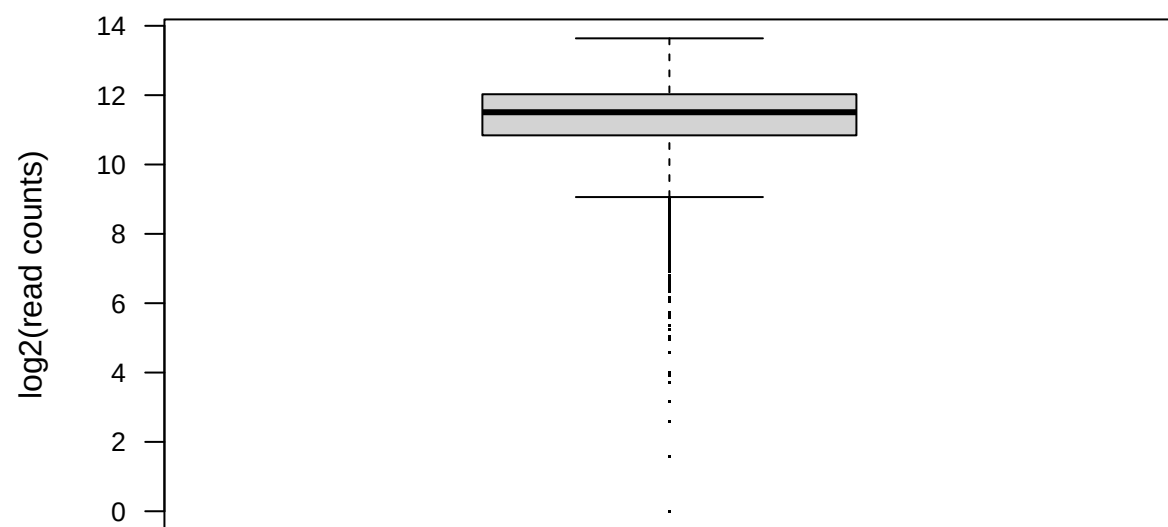
File	Label	NegSelQC	NegSelQCPval	NegSelQCPvalPermutation	NegSelQCPvalPermutationFDR	NegSelQCGene
21062D-05-01_S1_L001_R1_001.fastq	L1	0	1	1	1	0

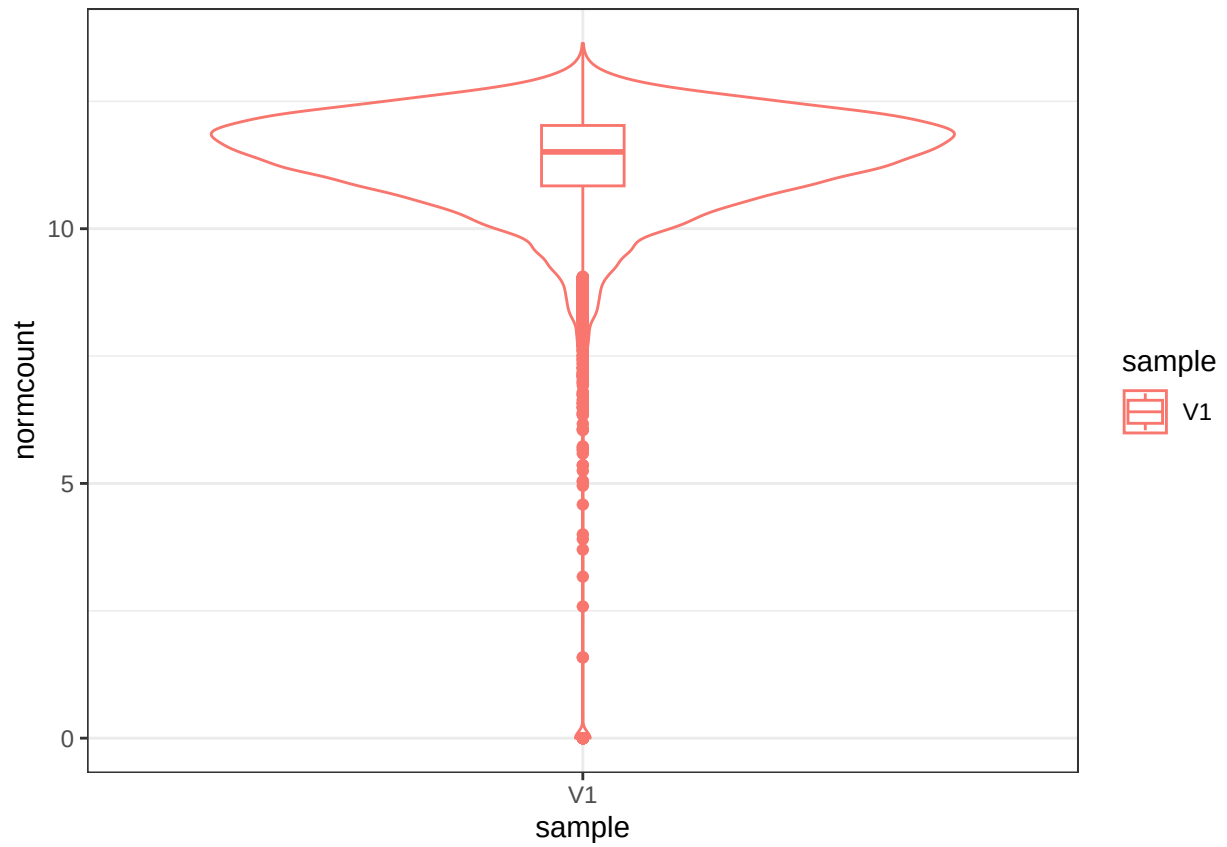
The meanings of the columns are as follows.

- NegSelQC: the enrichment score (ES) of essential genes in the negative selection list. The score is calculated using GSEA;
- NegSelQCPval: the associated p value of the enrichment score;
- NegSelQCPvalPermutation: the permuted p value of the enrichment score;
- NegSelQCPvalPermutationFDR: the adjusted permuted p value;
- NegSelQCGene: the number of genes used for the analysis.

Normalized read count distribution of all samples

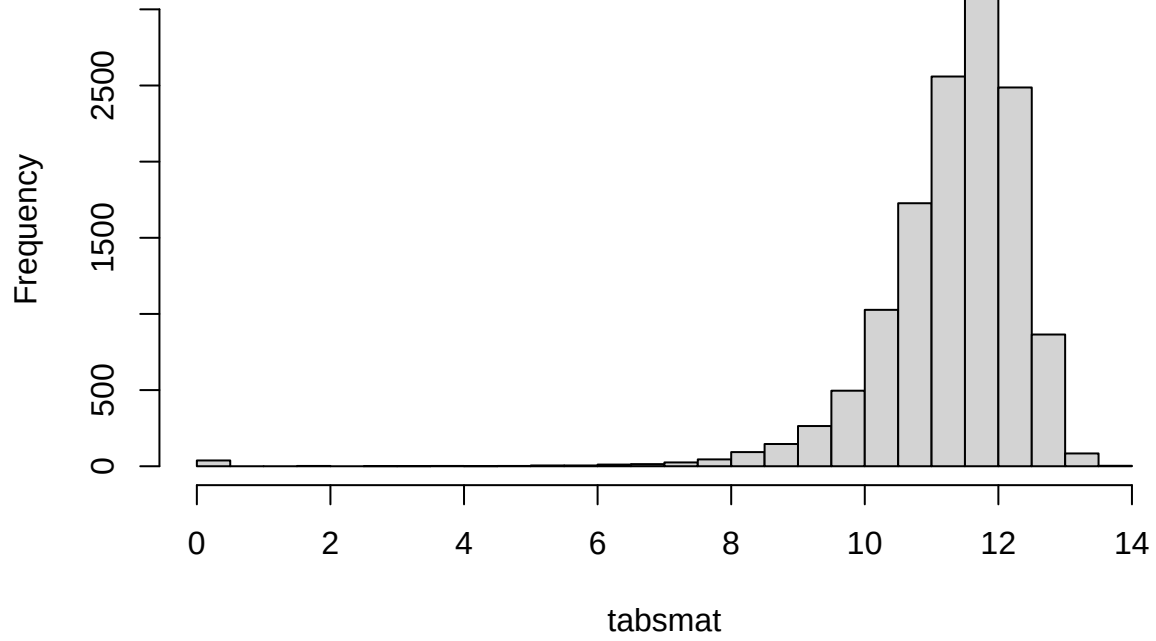
The following figure shows the distribution of median-normalized read counts in all samples.



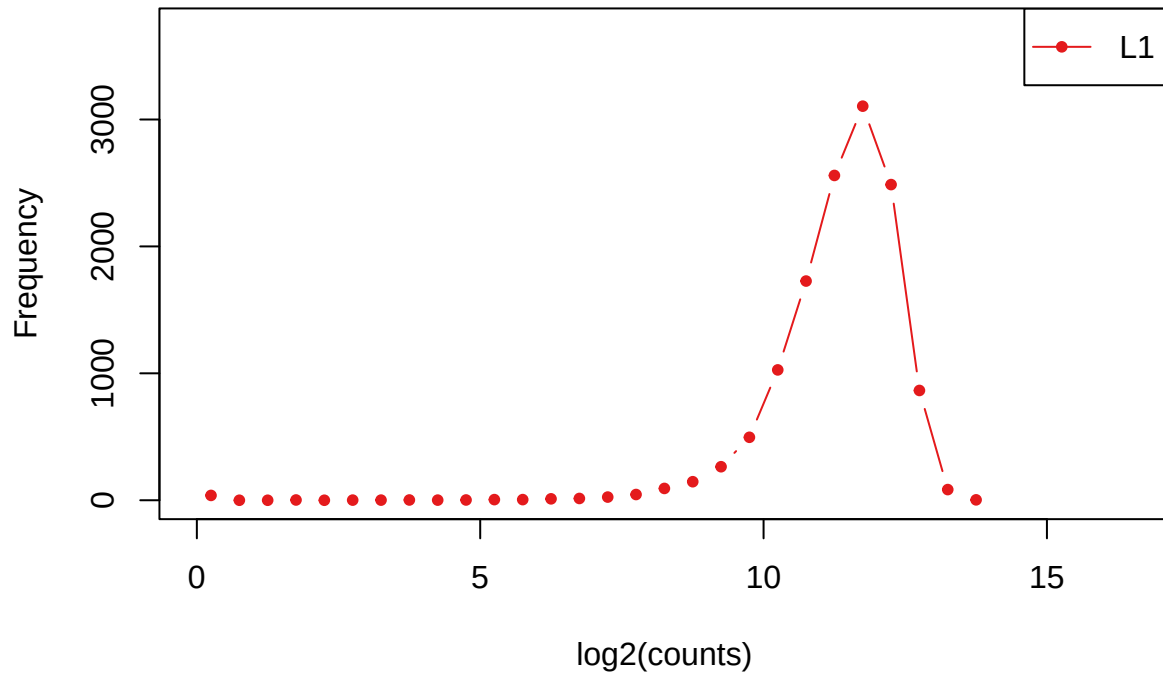


The following figure shows the histogram of median-normalized read counts in all samples.

Histogram of tabsmat



Distribution of read counts



Principle Component Analysis

The following figure shows the first 2 principle components (PCs) from the Principle Component Analysis (PCA), and the percentage of variances explained by the top PCs.

The variance of the PCs

```
## [1] "Skip PCA plot as there is only one sample."
```

```
## [1] 0
```

Sample clustering

The following figure shows the sample clustering result.

```
## [1] "Skip Clustering plot as there is only one sample."
```